

pH Rainbow

A Colorful Chemistry Adventure!  



BECOME A KITCHEN LAB DETECTIVE



Hello Future Scientists! 

Have you ever seen a magic trick performed by vegetables? Welcome to the **pH Rainbow Course Brief!** This guide is packed with fun games, incredible colors, and kitchen mysteries. We are going to discover how common foods and cleaners belong to secret invisible families that change colors right before your eyes!

Target Grade Level: Kindergarten to Grade 2 (Ages 5-8) 

Course ID: PLANRS-SCI-GRK2-004 | **Date:** June 2026

Designed by Planrs Academy. Perfect for Live Interactive Lab Sessions.

1. Welcome to the Color Science Show!

Science is all around us, especially inside our kitchens! The **pH Rainbow** chemistry session is an adventure made specifically for early childhood learners. Instead of using big, boring scientific terms, we turn chemistry elements into fun characters that kids can understand through tasting, feeling, and looking at bright color changes!

WHAT OUR LITTLE DETECTIVES WILL LEARN:

By using simple household materials like red cabbage and turmeric spice, students develop major cognitive frameworks and school readiness markers:

- **Spotting Differences:** Sorting foods and solutions into special families based on their taste, texture, and magical color outcome.
- **Hand & Muscle Skills:** Using plastic pipettes and droppers to pull and push liquids precisely into test spots, strengthening tiny hand muscles.
- **Real-World Connections:** Figuring out why everyday things happen—like why a yellow food stain on a shirt suddenly changes color when you wash it!



Planrs Parent & Teacher Note

We do not expect young children to understand complicated topics like "hydrogen ion count." Instead, we build permanent memory links by pairing sensory descriptions (sour, bitter, slippery, sharp) directly with physical color transitions!

TEACHER GUIDE: SETTING UP THE MINDSET

Teacher: "Put on your invisible lab glasses, detectives! (Pretend to slide glasses on). Today, our classroom is a high-tech science laboratory. We are going to test mysteries, search for clues, and make colors pop!"

2. Lab Detective Superpower: Safe Science!

Every great scientist has a superpower, and our primary superpower is **Safety First!** Before any dropper is lifted or any cup is mixed, every child must learn how to protect themselves in a lab space.

THE 'NO-SIP, NO-SNIFF' COMMAND RULE

Even though our chemicals today are made from normal food items from the kitchen, scientists never put unknown things near their noses or mouths unless given direct adult permission.

THE LAB DETECTIVE SAFETY BADGE

"If it is in a lab space, we **NEVER** sip!

We **NEVER** sniff!

We **ALWAYS** ask an adult first!"

HOW TO CHECK FOR SAFETY ALIGNMENT:

Detective Behavior	What it Means	Points Awarded
Hands at Rest	Keeping hands folded until the teacher says the activation phrase.	 10 Star Points
Pipette Control	Squeezing the dropper bubble gently without splashing anything outside the cup.	 10 Star Points
Adult Permission Check	Asking loudly: "Can I test this now, Lab Captain?" before mixing.	 10 Star Points

VERBATIM ACTION SCRIPT: SAFETY PLEDGE

Teacher: "Cross your arms like a brave superhero and repeat after me: *If it's in a lab space, we never sip, we never sniff, we always ask an adult!* Wow, you sound ready to protect your lab bench!"

3. The 5 Steps of Our Science Expedition

Our lab session is broken up into five fast, action-packed parts so that kids stay energized, engaged, and never get bored!

THE LAB JOURNEY ROADMAP

Journey Phase	Time Spent	What Cool Things Do We Do?	Teacher's Secret Mission
Step 1: The Hook	10 Minutes	We investigate an accidental food stain mystery from a yummy lunch box!	Make kids super surprised and excited.
Step 2: The Concept	15 Minutes	Meet the fictional Acid Family and Base Family characters.	Teach the core character traits clearly.
Step 3: Live Lab	25 Minutes	The big color show! Students drop indicator juice into sample cups.	Help kids manage their droppers cleanly.
Step 4: Synthesis	15 Minutes	Solve our Detective Quiz game together as a team.	Check how well they remember the colors.
Step 5: Outro	10 Minutes	Get our special take-home secret mission assignment!	Hand out home logs safely.

Adaptive Track: For older Grade 2 students, we can extend the lab by adding a third mystery category called the **Neutrals Family**. They are the peacekeeper liquids (like plain water) that don't change color at all!

4. Phase 1: The Mystery Aloo Paratha Stain!

Our adventure begins with a real crime scene from the lunchroom! Imagine you just finished eating a hot, buttery, delicious piece of **Aloo Paratha**. You look down, and gasp! There is a bright yellow stain right on your clean shirt!

THE SHOCKING COLOR SHIFT

What spice makes our food look yellow? That's right, it's **Turmeric**! But look at what happens next. When we run to the sink and rub the yellow stain with normal bathroom soap and water, it doesn't wash away. It instantly transforms into a deep, dark, mysterious **RUST-RED** spot!



THE DETECTIVE PUZZLE QUESTIONS

- How can a bright yellow spice turn red just by touching soap?
- Is it a magic spell, or is it cool kitchen science?
- Can we use this color code to unlock secrets about other liquids?

VERBATIM NARRATION SCRIPT: THE STAIN MYSTERY

Teacher: "Look closely at this yellow spot on my slide! It's bright like a little yellow sun. Now, let's touch it with some soap. *Click slide* Wow! Did it disappear? No way! It turned into a secret RED color! How did that happen? Let's travel to two invisible kingdoms inside our kitchen to find out!"

5. Phase 2: Meet Kingdom Acid! 🍊

Let's cross the bridge into our first invisible territory. Say hello to the **Acid Family!** They are the ultimate **Sour Flavor Heroes** of the kitchen world.

CHARACTER PROFILES: THE SOUR HEROES

- **Lucy Lemon:** She is bright, bursting with high energy, super sharp, and incredibly sour! She loves hanging out inside juicy limes, oranges, and lemons.
- **Tamarind Water:** He is calm but carries a strong, tangy punch that makes your lips puckered up.

THEIR SECRET COLOR SPELL CODE:

The Acid Family has a favorite magical liquid called **Purple Cabbage Juice**. Whenever they touch it, they cast an instant color spell:

🦁 **ACID + PURPLE JUICE = BRIGHT, FLAMING PINK-MAGENTA!** 🌸

👄 SENSORY CLUE CONNECTION

Whenever you bite into a sour fruit and your face automatically squeezes together tight like this—**make a funny sour face**—that is the Acid Family waving hello to your tongue! Children link that sharp feeling directly with hot pink tones.

VERBATIM DIALOGUE SCRIPT: KINGDOM ACID

Teacher: "Let's open the big wooden gates to Kingdom Acid! Meet Lucy Lemon! She loves bright, energetic styles. If you give her our deep purple magic vegetable juice, she waves her scientific wand and—POW!—turns it into a beautiful, glowing hot pink! Can you make a sour face like Lucy Lemon?"

6. Phase 2: Meet Kingdom Base!

Now, let's cross the slow river to visit their next-door neighbors. Wave hello to the **Base Family**! They are completely different from the Acid Family. They are the **Smooth, Slippery Friends** of our household.

CHARACTER PROFILES: THE SMOOTH FRIENDS

- **Sammy Soap:** He loves making giant bubbles, feels super slippery to touch, and slides smoothly across your hands. (He tastes bitter, but we never eat him!).
- **Baking Soda Water:** She is soft, calm, looks like a white chalky powder, and loves cleaning up mess zones.

THEIR SECRET COLOR SPELL CODE:

The Base Family absolutely hates pink! They prefer relaxing, cool tones from the deep ocean. Look at what they do to our magical juice:



BASE + PURPLE JUICE = COOL OCEAN BLUE / EMERALD GREEN! 



TOUCH CLUE CONNECTION

Bases are defined by how they feel. When you wash your hands and the soap slides easily over your skin, that's the feeling of a base! This physical texture helps kids remember cleaners and detergents easily.

VERBATIM DIALOGUE SCRIPT: KINGDOM BASE

Teacher: "Now let's meet Sammy Soap and his friends! They are calm and smooth. Rub your hands together like you are washing them. See how slippery that feels? When Sammy Soap mixes with our purple magic juice, it turns into an incredible, shimmering emerald green-blue sea color!"

7. Phase 3: Setting Up the Live Lab Bench

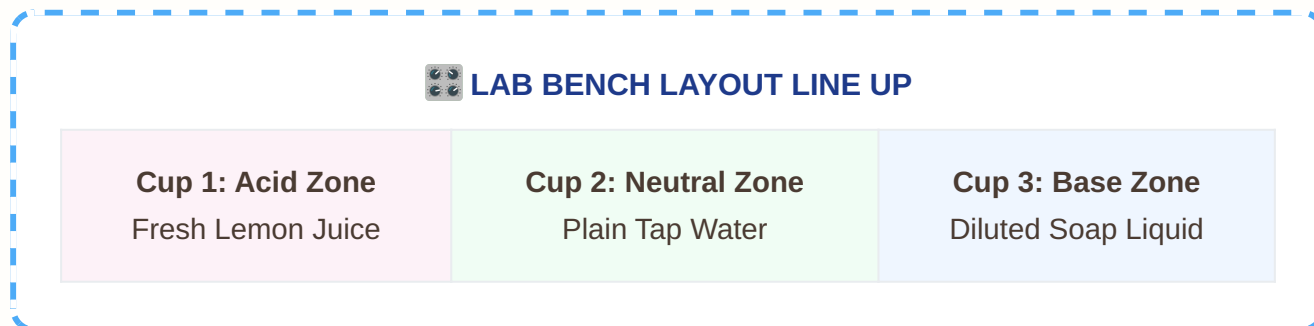
To run our great color show successfully, teachers and caregivers should prepare the non-toxic lab reagents 30 to 45 minutes before starting. Everything is food-grade and perfectly safe!

1. MAKING THE "MAGIC INDICATOR JUICES"

- **Red Cabbage Fluid (The Purple King):** Chop up 1 cup of red cabbage, soak it in hot boiling water for 15 minutes until the water becomes an intense, dark purple color. Filter out the vegetable chunks and let it cool.
- **Turmeric Solution (The Yellow Alarm):** Stir 1 tablespoon of organic yellow turmeric powder into 100ml of warm water. Mix it well until it looks like bright liquid gold.

2. GATHERING OUR TESTING TARGETS

Lined up across each student's desk should be three transparent sample cups from left to right:



MATERIAL SAFETY INVENTORY CONTROL TABLE

Item Description	Amount Needed	Safety Rating	Clean Up
Purple Cabbage Juice	30 mL per kid	😊 Totally Safe / Edible	Pour down sink
Yellow Turmeric Fluid	20 mL per kid	⚠️ Can stain fabrics	Pour down sink
Diluted Soap Liquid	15 mL per kid	👁️ Eye Sting Hazard	Pour down sink
Plastic Pipettes (3ml)	3 Units	💪 Plastic / Reusable	Wash with water

8. Phase 3: The Big pH Rainbow Color Show!

Now comes the most exciting part! It is time to run the live lab experiment. Follow these four guided steps carefully to create your rainbow columns.

THE STEP-BY-STEP SCIENCE PROCEDURE:

- 1. The Inspection:** Look closely at the big central bowl of your Magic Juice. Shout out what color it is natively. (*Answer: Deep Purple!*)
- 2. The Acid Test:** Dip your plastic pipette into the purple juice. Squeeze the top bubble to suck some up. Move it carefully over **Cup 1 (Lemon Juice)**. Push 5 drops down. Swirl gently. Watch it instantly change to **Vibrant Pink!**
- 3. The Base Test:** Refill your pipette with purple juice. Move it over **Cup 3 (Soap Solution)**. Let 5 drops fall. Swirl gently. Watch it drift into a deep **Emerald Green-Blue!**
- 4. The Control Balance Test:** Drop 5 drops of purple juice into **Cup 2 (Plain Water)**. Look closely. Did it change? No! It stays **Deep Purple** because water is balanced and neutral!

OUR OFFICIAL RAINBOW RESULT MATRIX

Testing Cup	Starting Color	What We Added	Final Rainbow Color
Cup 1: Lemon Juice	Clear / Pale Yellow	Purple Magic Juice	 Bright Pink-Magenta
Cup 2: Plain Water	Clear	Purple Magic Juice	 Deep Purple (No Change)
Cup 3: Soap Solution	Clear / Milky White	Purple Magic Juice	 Emerald Green / Blue

LIVE LAB FLOW SCRIPT

Teacher: "Pipettes up, scientists! Squeeze, dip, and release. Now let 5 drops fall into Cup 1... tap, tap, tap, tap, tap. Swirl it! Wow, look at that gorgeous pink splash! You just triggered Acid Magic!"

9. Decoding the Rainbow's Secret Formula

Let's take a deep breath, step back from our lab benches, and look at our amazing results. We have unlocked a secret color language that nature uses to communicate internal molecular states!

THE RAINBOW'S SECRET CODE CARD

-  **Red & Pink Mean Sour Acids:** Whenever our purple indicator juice shifts to pink, it tells us that a fast, sharp sour acid family member is hiding inside that cup.
-  **Green & Blue Mean Smooth Bases:** Whenever our indicator liquid shifts to cool green or blue, it alerts us that a slippery, cleaning base family member is present.
-  **Purple Means Balanced Neutrals:** If the liquid stays its original purple color, it means it is a peaceful substance like pure tap water that doesn't side with either family!

DEEP BRAIN DETECTIVE QUESTIONS

Question 1: Can a kitchen liquid be super sour and super slippery at the exact same time?

Target Logic Response: No way! They are opposite chemical families. If you mix them together, they fight and cancel each other out! This is called neutralization.

Question 2: Why does the color change instead of just getting watered down?

Target Logic Response: Because the tiny hidden particles inside the lemon juice or soap actively change the shape of the purple cabbage juice molecules!

Planrs Educational Breakthrough: By teaching young minds that color is an expression of an internal state rather than just an arbitrary design, we prepare them natively for advanced chemistry later in school!

10. Phase 4: Detective Quiz Game - Case #1

It's time to put your new science powers to the test! We have two tough mystery cases to solve. Let's see if you can crack them using our color code cards.

THE MYSTERY OF THE SOUR CURD (YOGURT) 🥛

Problem Statement: Detective Elephant is standing in the kitchen holding a dropper full of purple Magic Juice over a bowl of thick, creamy, white **Sour Curd**. What color will the curd turn when the purple drops splash into it?



CHOOSE YOUR ANSWER CARD:

- **Option A: Bright Pink / Magenta Bowl** 🌸
✓ **CORRECT ANSWER!** Why? Because the prompt uses the magic keyword "**SOUR.**" Sour belongs directly to the Acid Family profile, and acids always turn purple juice into bright pink! (Curd contains lactic acid).
- **Option B: Emerald Green Ocean** 🌲
✗ **INCORRECT.** Green means it is a smooth, slippery base. Curd isn't slippery like soap water!
- **Option C: Plain Unchanged Purple** 🌌
✗ **INCORRECT.** Purple means it is plain water. Curd has a strong sour punch and will definitely change the color.

TEACHER GUIDANCE PIVOT LOOP

Teacher (If students guess Option B): "Wait a minute, detectives! Look past the color white. Put it on your imaginary tongue—does plain curd taste bitter like soap, or does it taste sharp and tangy like a lemon?"

Students: "Tangy and sharp!"

Teacher: "Which means it belongs to...?"

Students: "THE ACID FAMILY! Option A!" 🌟

11. Detective Quiz Game - Case #2 & Stain Solution

Let's solve our second case and finally unlock the huge secret behind our accidental lunch stain mystery!

QUIZ CASE #2: THE SOAP WATER REVIEW 🧼

Problem Statement: A lab assistant hands you a cup filled with clear, slippery soap water. If you drop our purple cabbage juice into it, what color option will appear?

- **Option A: Pink-Red Spheres** 🌸 (Incorrect—soap is not a sour acid).
- **Option B: Cool Sea Green / Ocean Blue** 🌊 (✓ **CORRECT!** Soap water is a classic base, turning cabbage indicator into beautiful blue-green).
- **Option C: Deep Purple** 🟣 (Incorrect—soap changes the indicator state heavily).

THE BIG ALOO PARATHA STAIN MYSTERY: SOLVED! 🔍

Now we have the ultimate scientific key to explain our lunch shirt! **Turmeric spice is a natural yellow indicator.** It acts like a silent chemical alarm system inside our food:

Liquid Added	Family Group	Turmeric Color Result	Real-World Sight
Lemon Juice	ACID FAMILY	Stays Bright Yellow	No change on stained cloth
Plain Tap Water	NEUTRAL FAMILY	Stays Bright Yellow	Washing with water alone
Soap Water	BASE FAMILY	☀️ Turns Rust-Red!	Wiping with bathroom soap

VERBATIM FINALE NARRATIVE SCRIPT

Teacher: "So when your yellow turmeric spot turned bright red upon touching soap, it was simply the turmeric spice yelling out loud: *'Hey! Help! I just found an invisible member of the Base Family!'* You solved the crime scene with chemistry!" 🎉

12. Phase 5: Take-Home Mission - The Yellow Code!

To seal in your lab superpowers, you are now being given a critical take-home experiential assignment called **The Yellow Code**. You get to be the Lead Recording Detective at home!



MANDATORY ADULT SUPERVISION LAWS

Because concentrated cleaning products can sting your eyes, a parent or adult guardian must handle all liquids and mix solutions. Students act as the **Lead Log Recording Detectives**, standing back safely and writing observations down!

HOME LAB ACTION INSTRUCTIONS:

1. **Prepare the Yellow Alarm Fluid:** Mix 1 teaspoon of turmeric powder into a small glass of warm water until it looks like beautiful liquid gold.
2. **Set Out the Targets:** Prepare two small clear cups. Fill **Cup A** with lemon juice or white kitchen vinegar. Fill **Cup B** with laundry soap or liquid hand soap mixed with water.
3. **Trigger the Reaction:** Add 1 tablespoon of your yellow turmeric water into Cup A. Look for changes. Next, add 1 tablespoon into Cup B. Watch the sudden pop to rust-red!
4. **The Advanced Reset Trick:** Squeeze a whole fresh lemon directly into the red liquid in Cup B. Watch as the high acid content overpowers the base, neutralizing it and resetting the color back to yellow!



MY STUDENT LAB OBSERVATION REPORT SHEET

Experiment Step	My Predicted Color	My Observed Color	Chemical Family
Turmeric Water + Lemon Juice	_____	_____	Acid Family 🍋
Turmeric Water + Soap Liquid	_____	_____	Base Family 🧼

13. Educational Quality Standards & Review Metrics

As our colorful chemistry session wraps up with a celebration sequence, educators utilize this standardized evaluation metric checklist to audit student understanding and session success benchmarks.

KEY POST-LAB REVIEW CHECKPOINTS:

- Ensure all students have stopped touching droppers, safely capped all sample cups, and washed their hands cleanly.
- Verify that every student can verbally link the character traits (Sour/Acid vs. Slippery/Base) to their respective indicator results.
- Distribute copies of the **Yellow Code Home Lab Sheet** to parents via alignment emails.

CURRICULUM EVALUATION BENCHMARK STANDARDS

Target Skill	★ Excellent (80% +)	👍 Developing (50-80%)	⚠️ Needs Work (<50%)
Safety Rule Recall	Instantly repeats "No-Sip, No-Sniff" loudly without any teacher help.	Remembers the rule only after looking at the slide badge.	Attempts to smell or taste materials during active lab tasks.
Color Property Mapping	Flawlessly links Pink to Acid and Green/Blue to Base outputs.	Mixes up the colors at first, but corrects quickly using character names.	Cannot match color outputs to underlying character properties.
Real-World Application	Clearly explains that soap water is a base that shifts turmeric red.	Identifies soap as a base but cannot explain the turmeric change.	Views the entire color shift as a magic trick without scientific logic.

"SUPERB JOB, LAB DETECTIVES! CLICK TO CLOSE SESSION!" 



Planrs Academy Curriculum Certification Statement: All narrative arcs, safety protocols, and motor-skill exercises align completely with international early elementary developmental milestones.

Primary Source Reference: PDF Resource: PH Rainbow slides-compressed.pdf (Slides 1-14 inclusive).